

Al Qadisiyah University
Collage of Medicine
Anatomy Dep.

Cardiovascular System

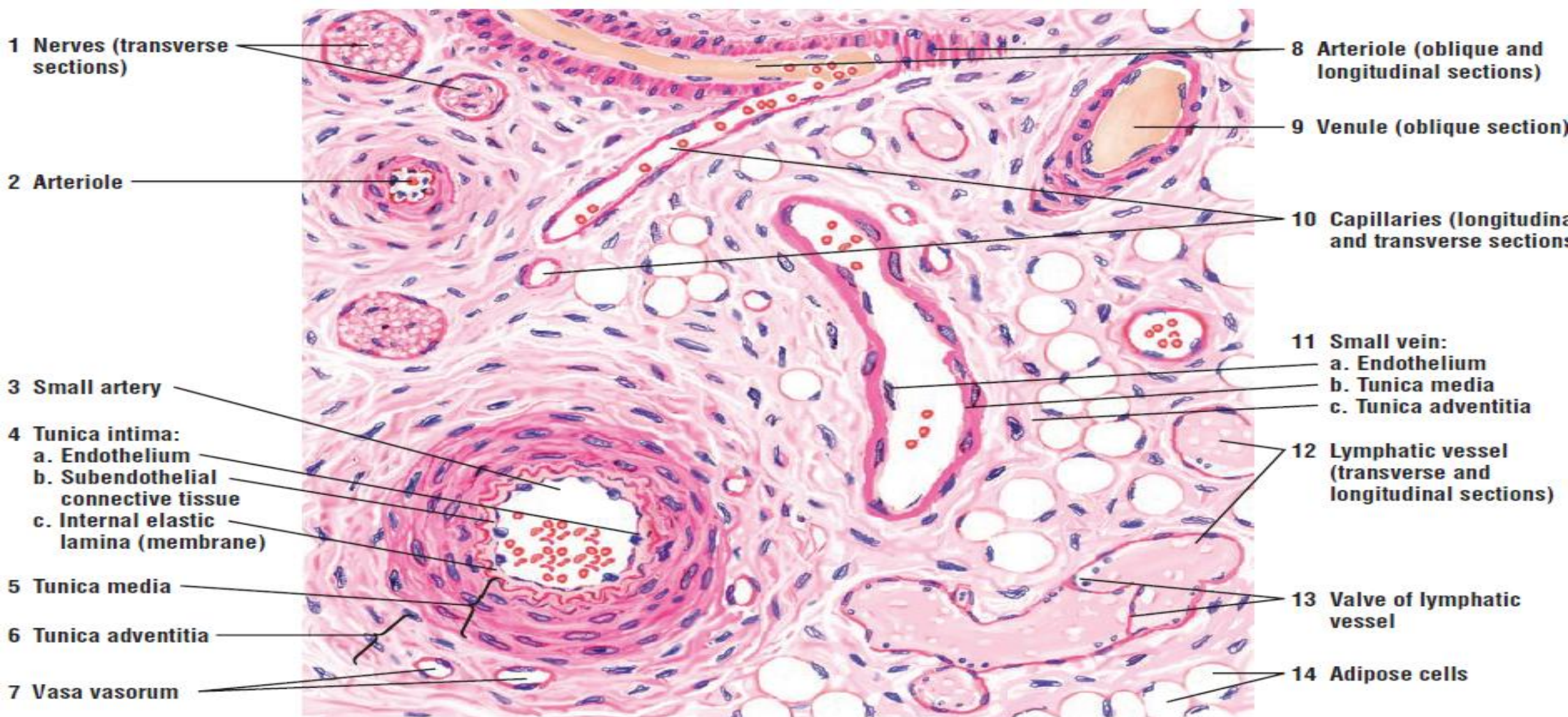


FIGURE 8.1 ■ Blood and lymphatic vessels in the connective tissue. Stain: hematoxylin and eosin. Low magnification.

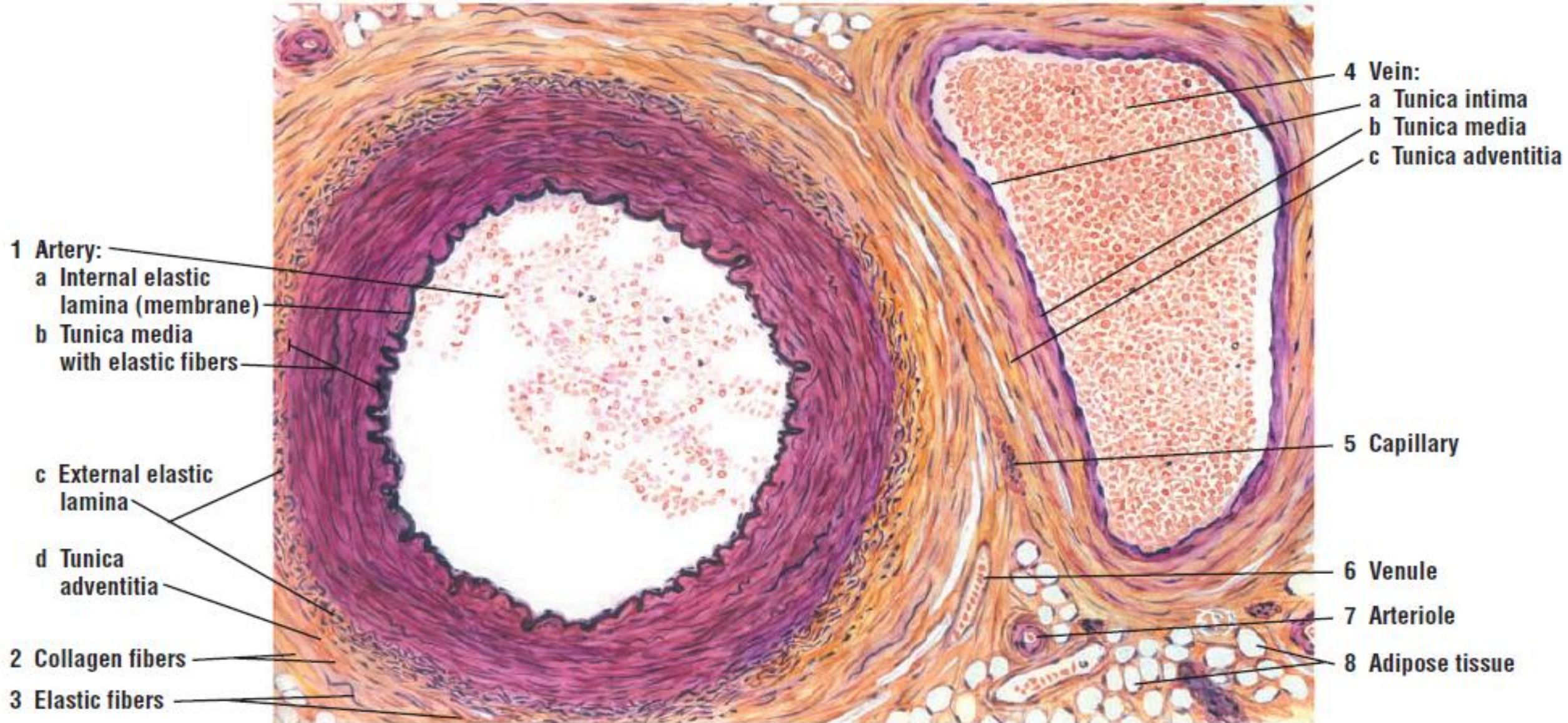


FIGURE 8.2 ■ Muscular artery and vein (transverse section). Stain: elastic stain. Low magnification.

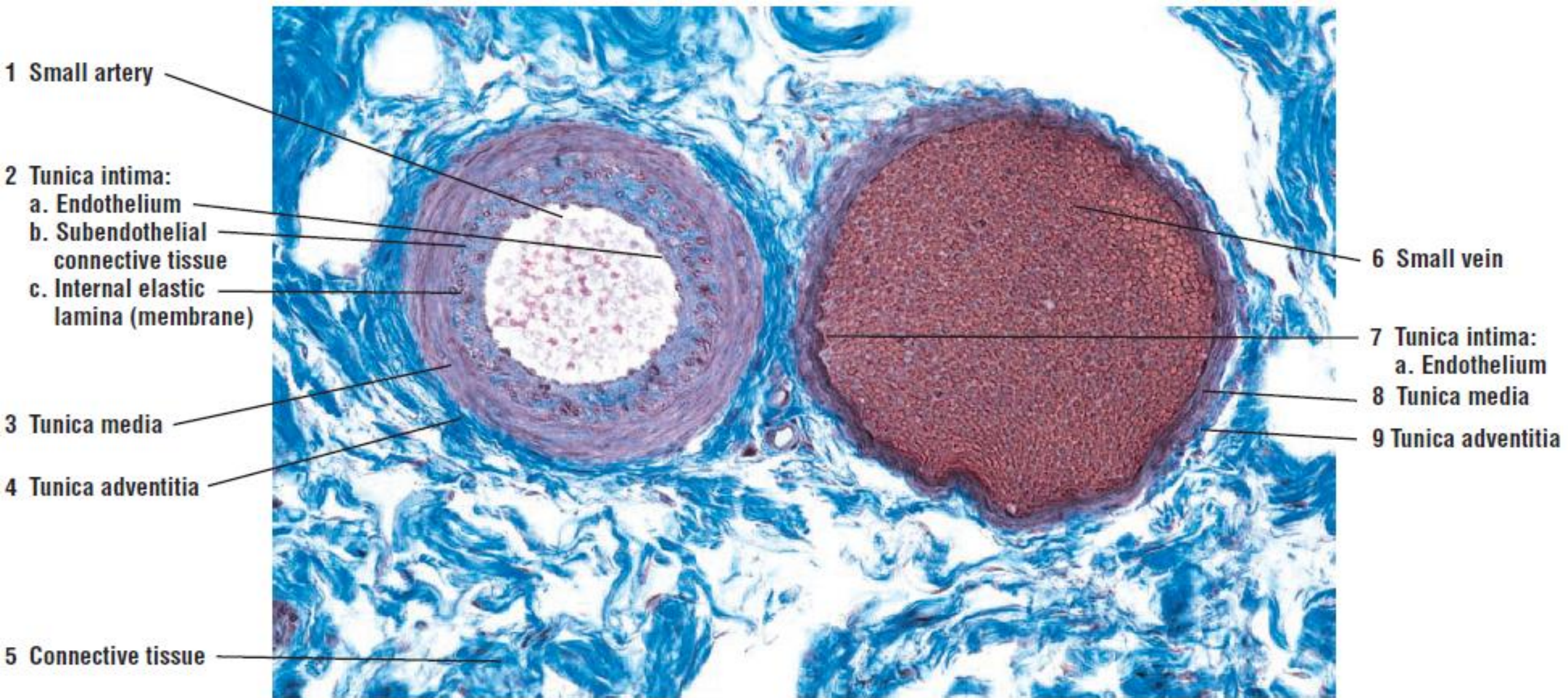


FIGURE 8.3 ■ Artery and vein in dense irregular connective tissue of vas deferens. Stain: iron hematoxylin and Alcian blue. $\times 64$.

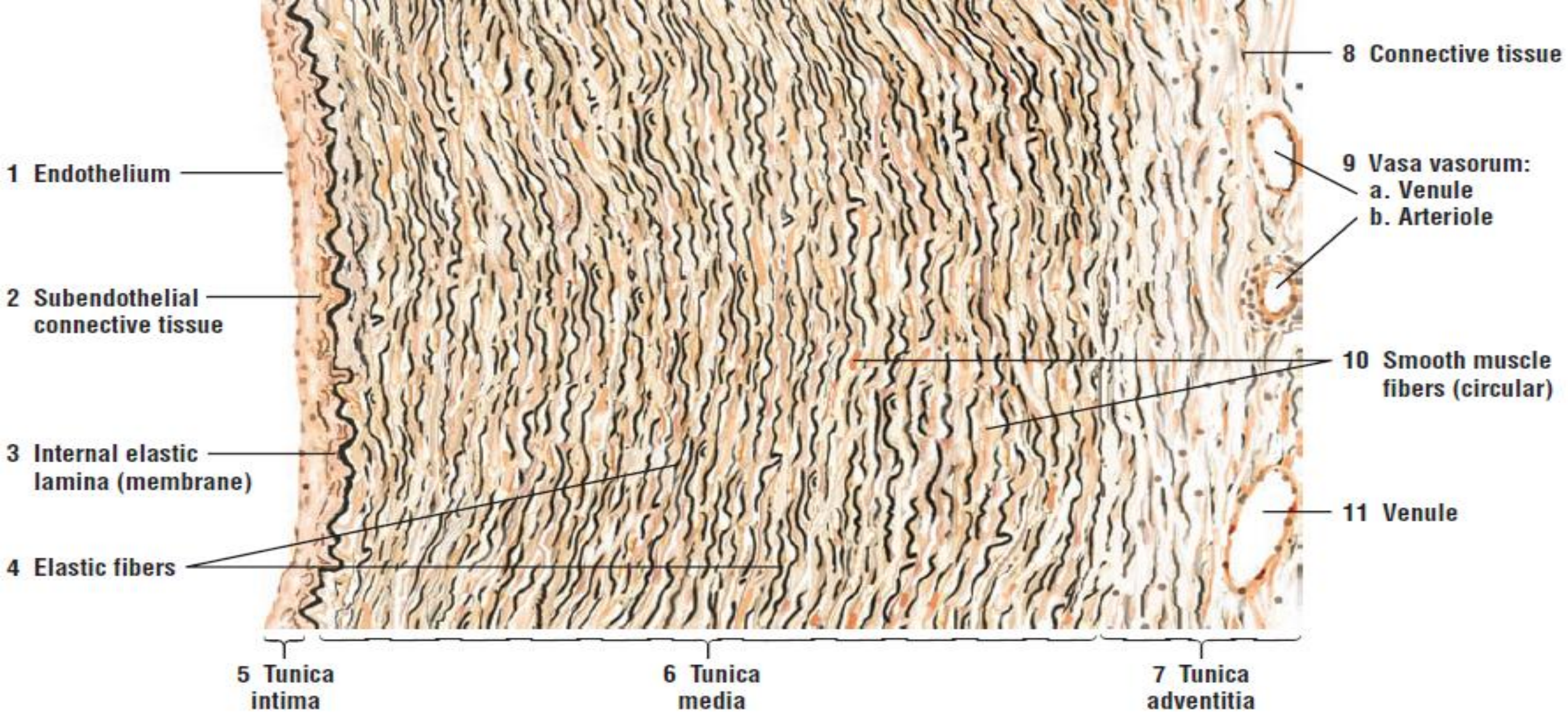


FIGURE 8.4 ■ Wall of a large elastic artery: aorta (transverse section). Stain: elastic stain. Low magnification.

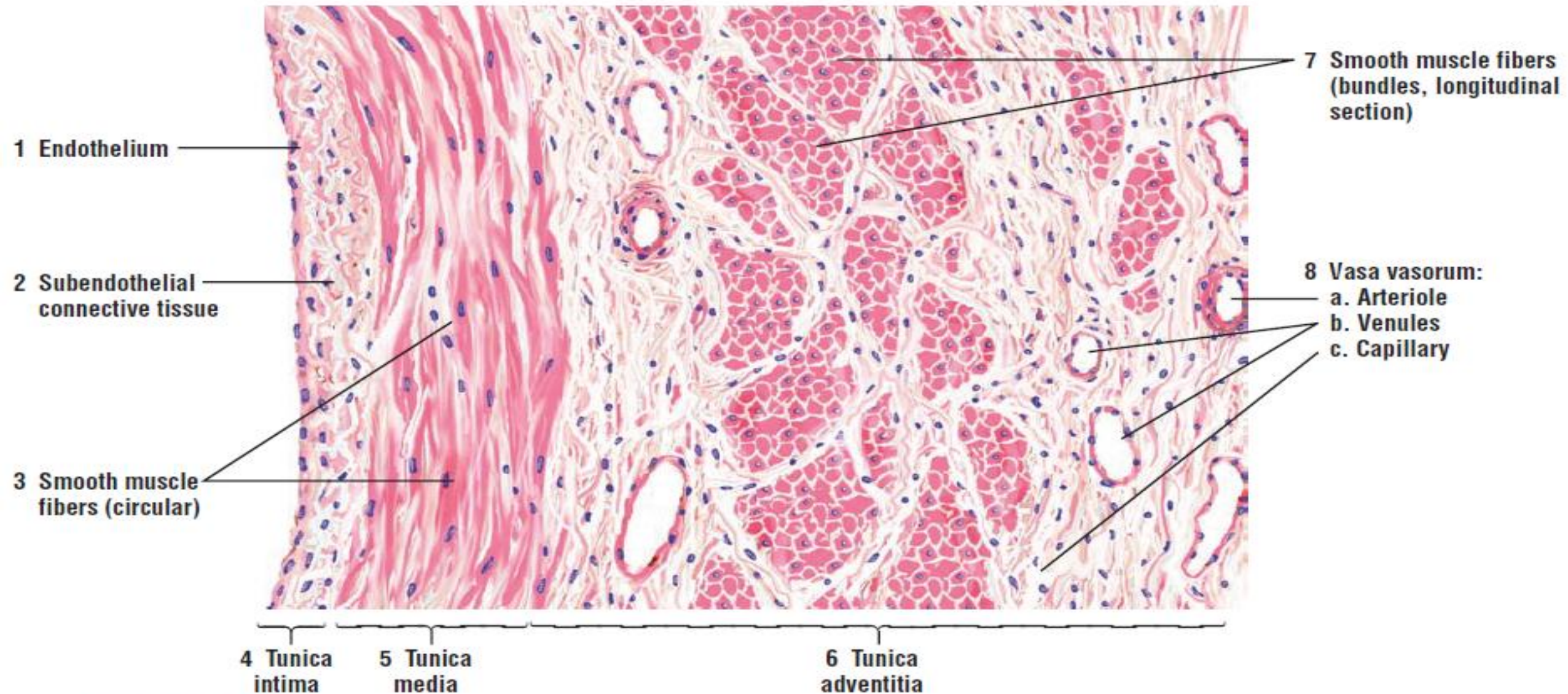
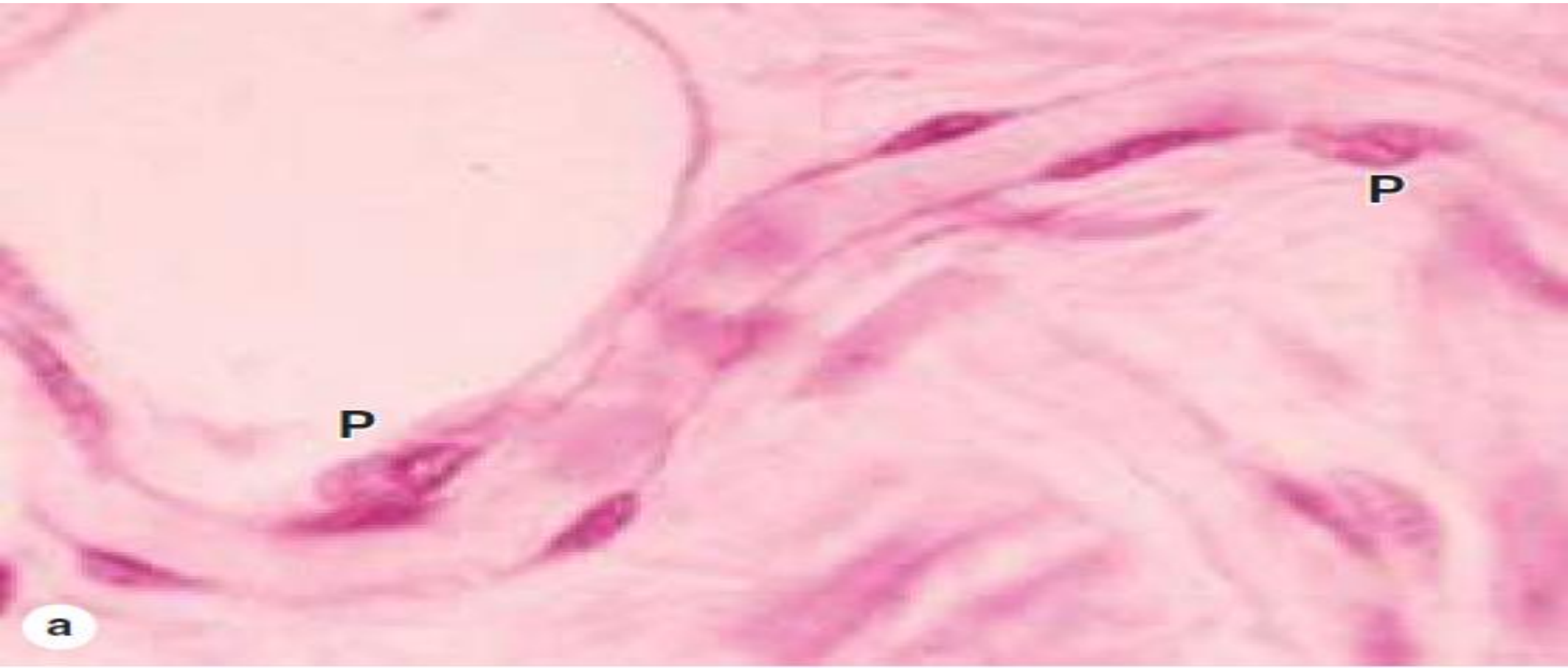
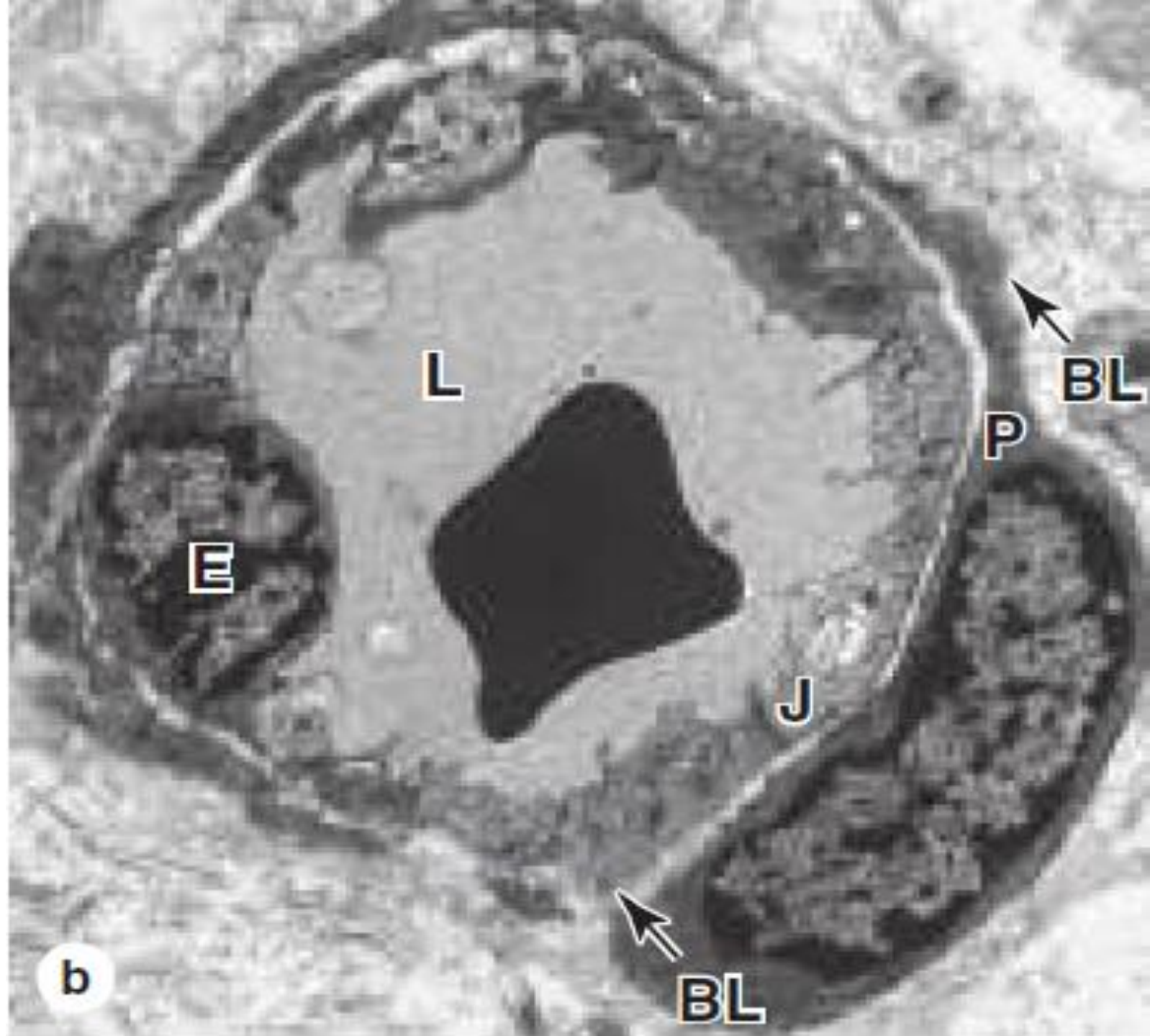


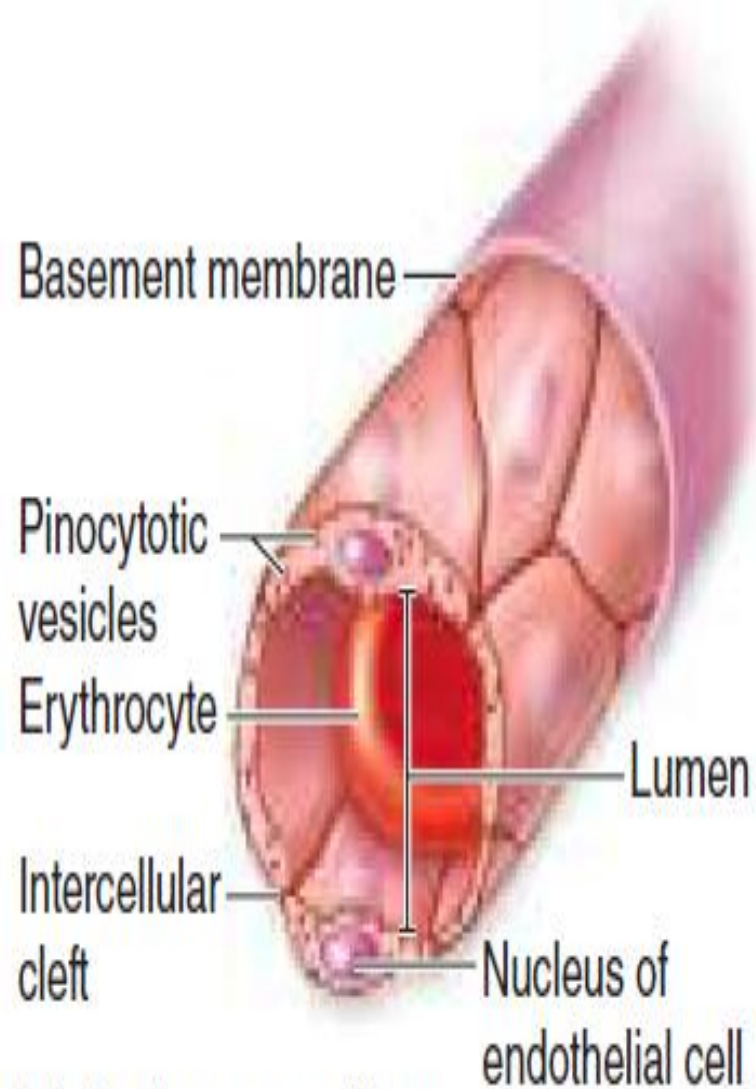
FIGURE 8.5 ■ Wall of a large vein: portal vein (transverse section). Stain: hematoxylin and eosin. Low magnification.

Capillaries With pericytes (P):
(X400; H&E of a spread mesentery preparation)

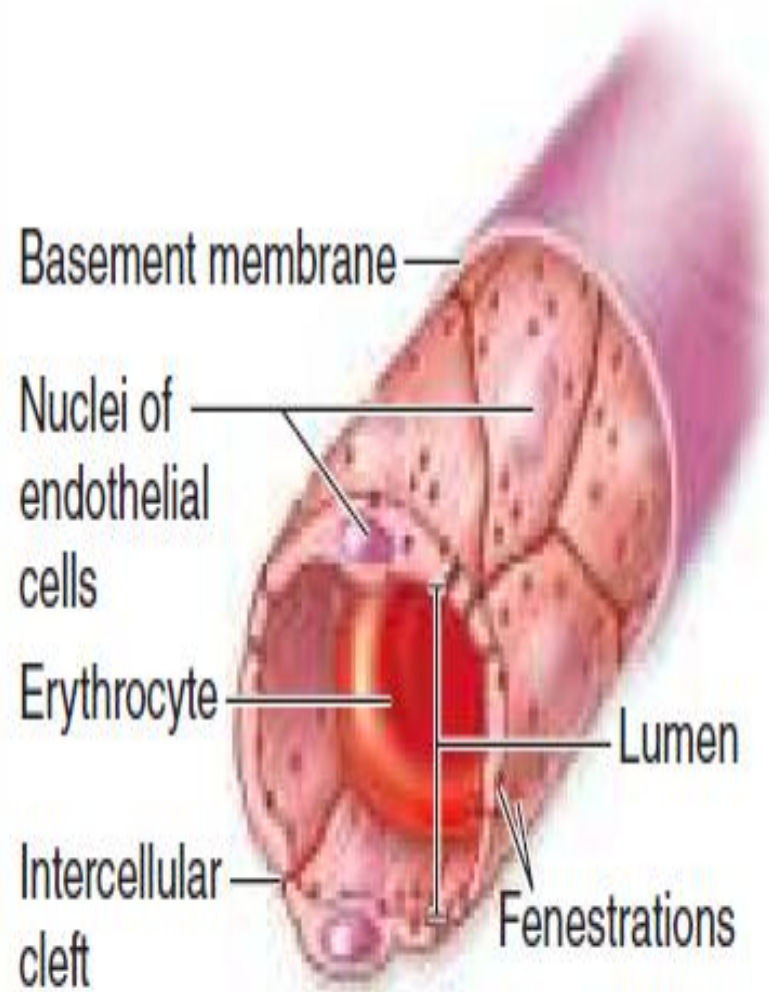


TEM of a capillary cut transversely, showing the nucleus of one thin capillary endothelial cell (E). Endothelial cells form the capillary lumen (L), are covered by a basal lamina (BL), and are bound tightly together with junctional complexes (J). One pericyte (P) is shown, surrounded by its own basal lamina (BL) and with cytoplasmic extensions which surround the endothelial cells. (X13,000)

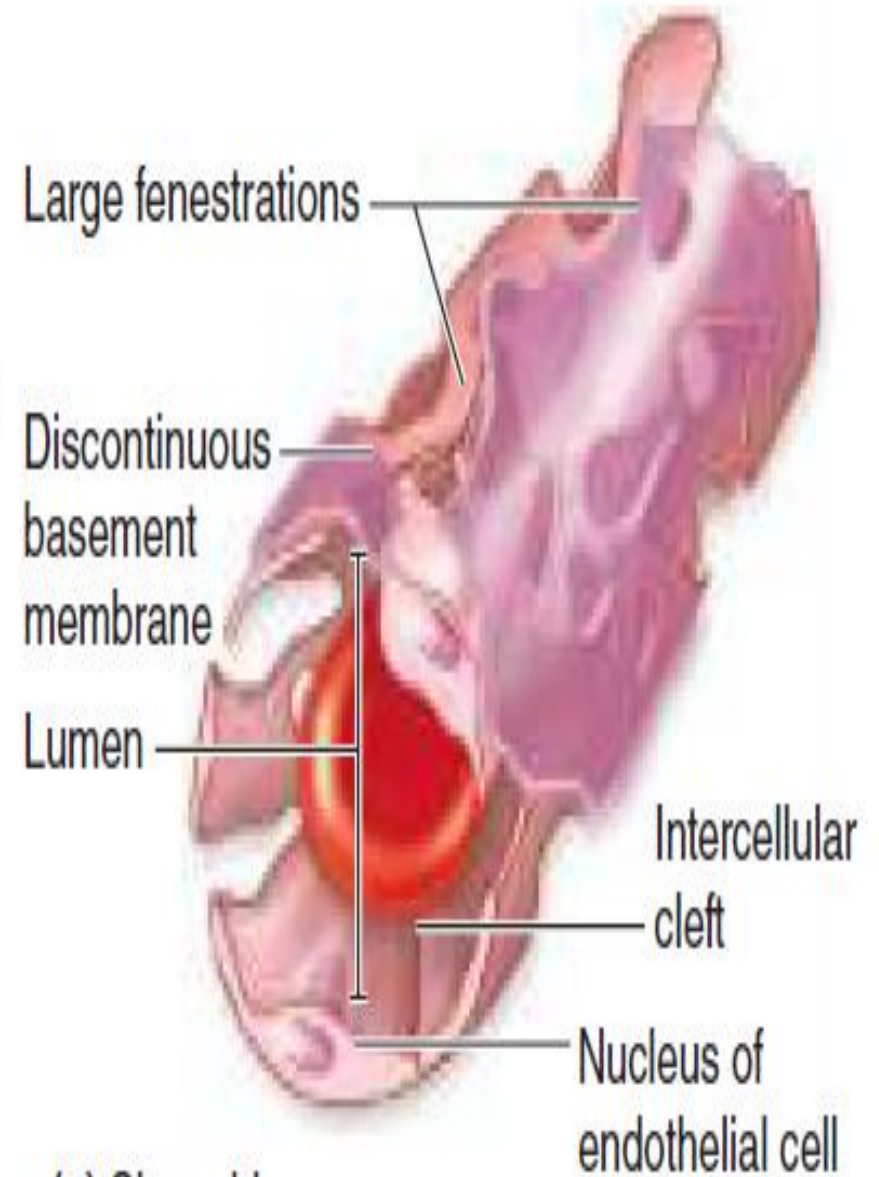




(a) Continuous capillary

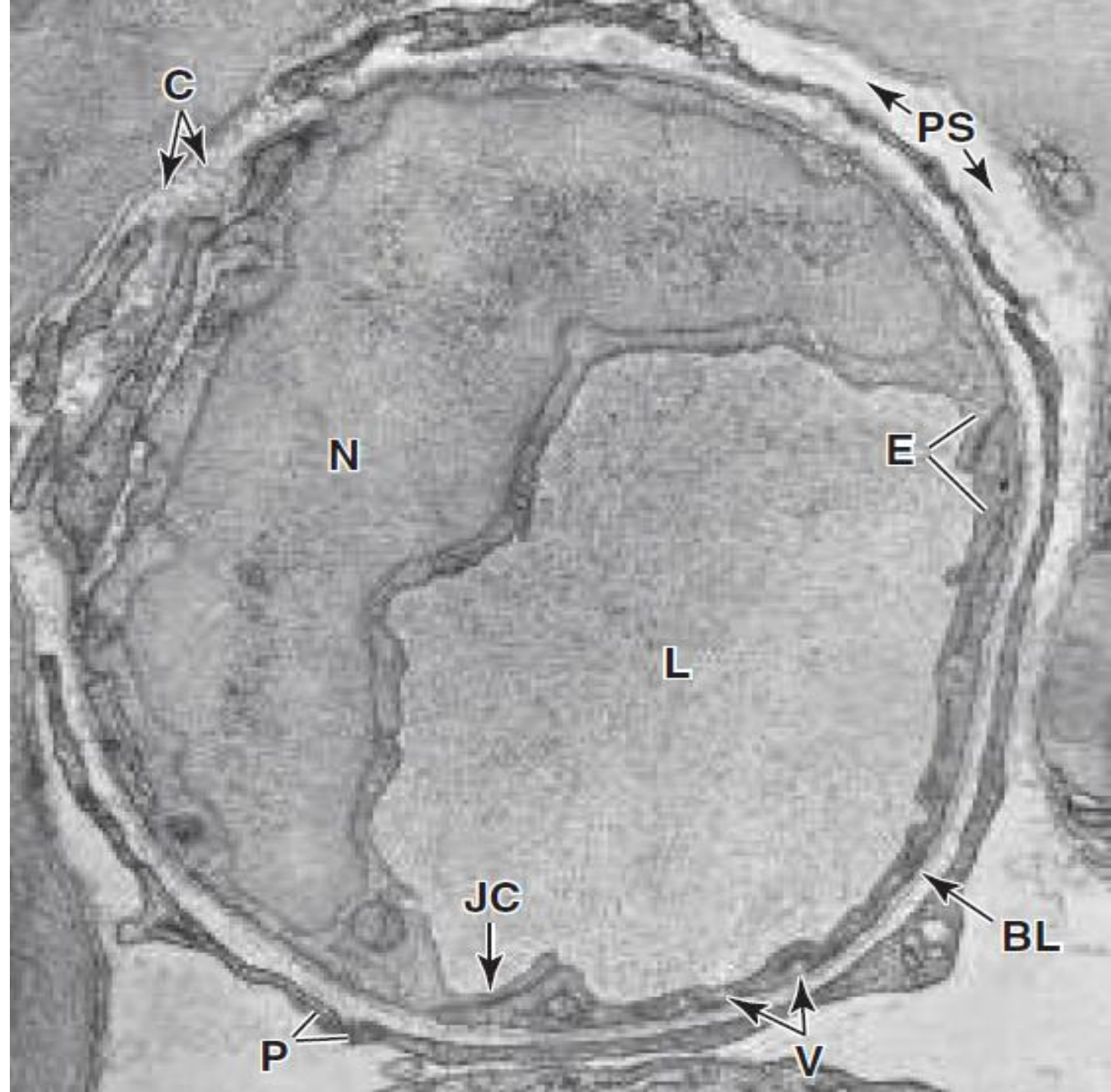


(b) Fenestrated capillary

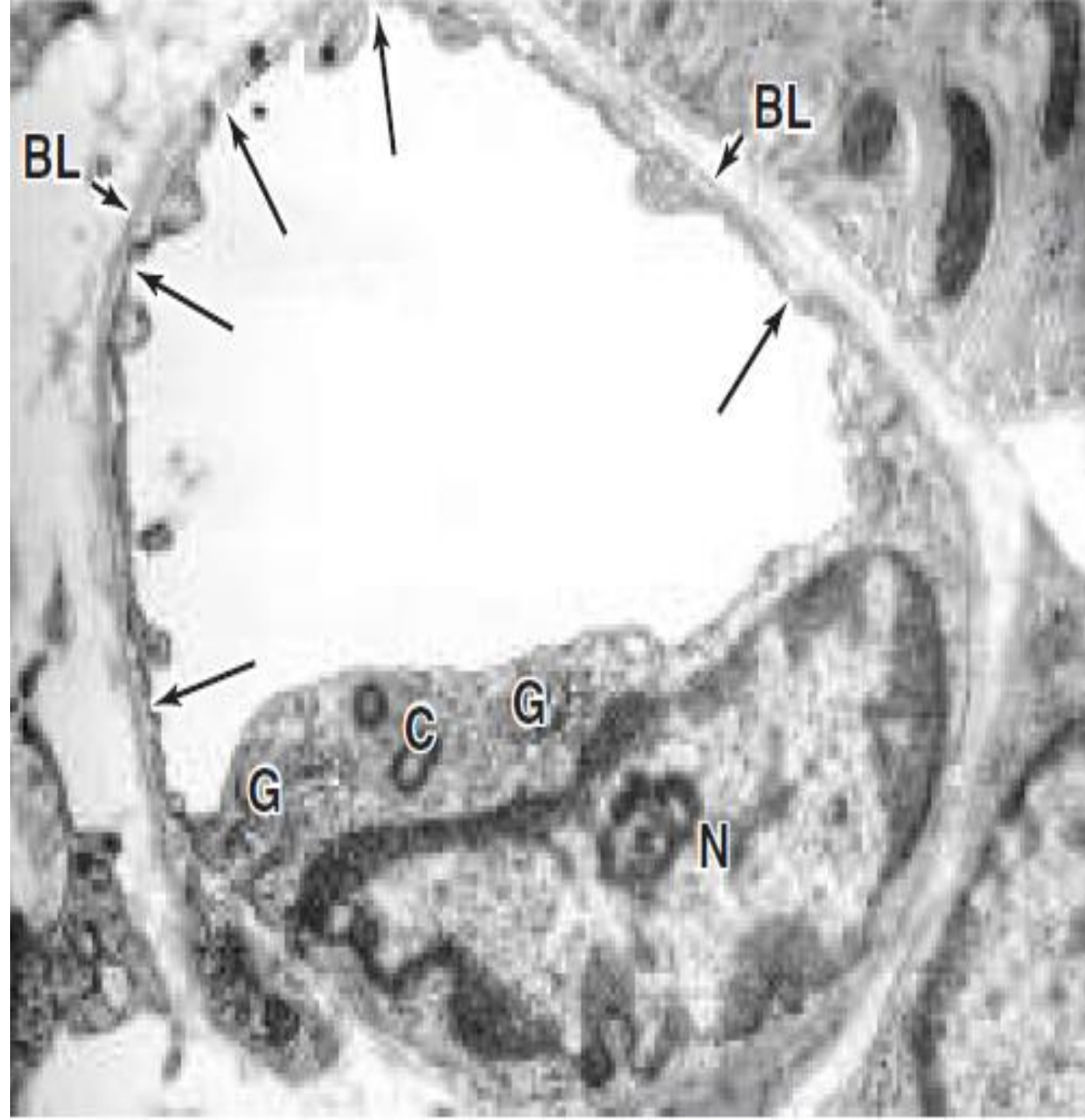


(c) Sinusoid

Continuous capillaries exert the tightest control over what molecules leave and enter the capillary lumen (L). The TEM shows a continuous capillary in transverse section. An endothelial cell nucleus (N) is prominent, and tight or occluding junctions are abundant in the junctional complexes (JC) at overlapping folds between the endothelial cells (E). Numerous transcytotic vesicles (V) are evident. All material that crosses continuous capillary endothelium must pass through the cells, usually by diffusion or transcytosis. Around the capillary are a basal lamina (BL) and thin cytoplasmic extensions from pericytes (P). Collagen fibers (C) and other extracellular material are present in the perivascular space (PS). (X10,000)



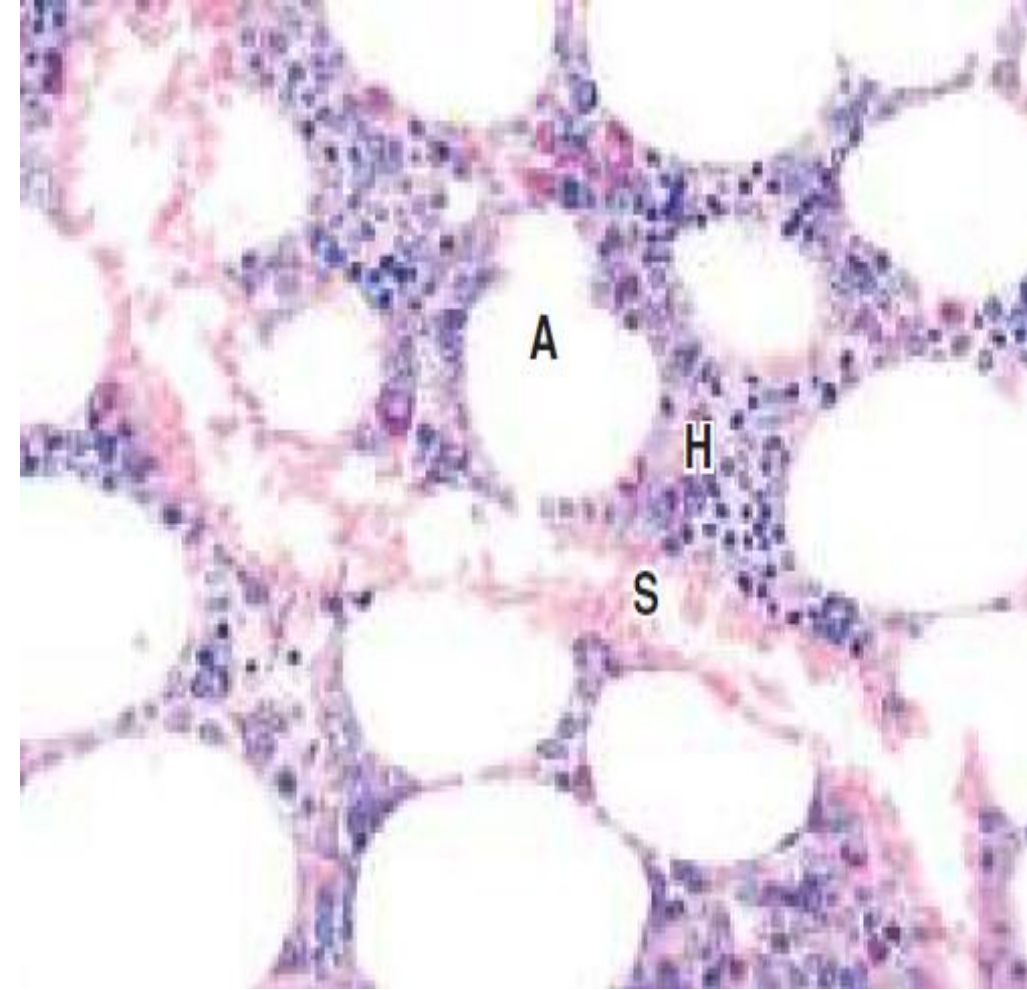
Fenestrated capillaries are specialized for uptake of molecules such as hormones in endocrine glands or for outflow of molecules such as in the kidney's filtration system. TEM of a transversely sectioned fenestrated capillary in the peritubular region of the kidney shows many typical fenestrae closed by diaphragms (arrows), with a continuous basal lamina surrounding the endothelial cell (BL). In this cell the Golgi apparatus (G), nucleus (N), and centrioles (C) can also be seen. Fenestrated capillaries allow a freer exchange of molecules than continuous capillaries and are found in the intestinal wall, kidneys, and endocrine glands. (X10,000)



Sinusoidal capillaries or sinusoids generally have much greater diameters than most capillaries and are specialized not only for maximal molecular exchange between blood and surrounding tissue but also for easy movement of blood cells across the endothelium.

The sinusoid (S) shown here is in bone marrow and is surrounded by tissue containing adipocytes (A) and masses of hematopoietic cells (H). The endothelial cells are very thin and cell nuclei are more difficult to find than in smaller capillaries.

Ultrastructurally sinusoidal capillaries are seen to have large fenestrations through the cells and large discontinuities between the cells and through the basal lamina. (X200; H&E)



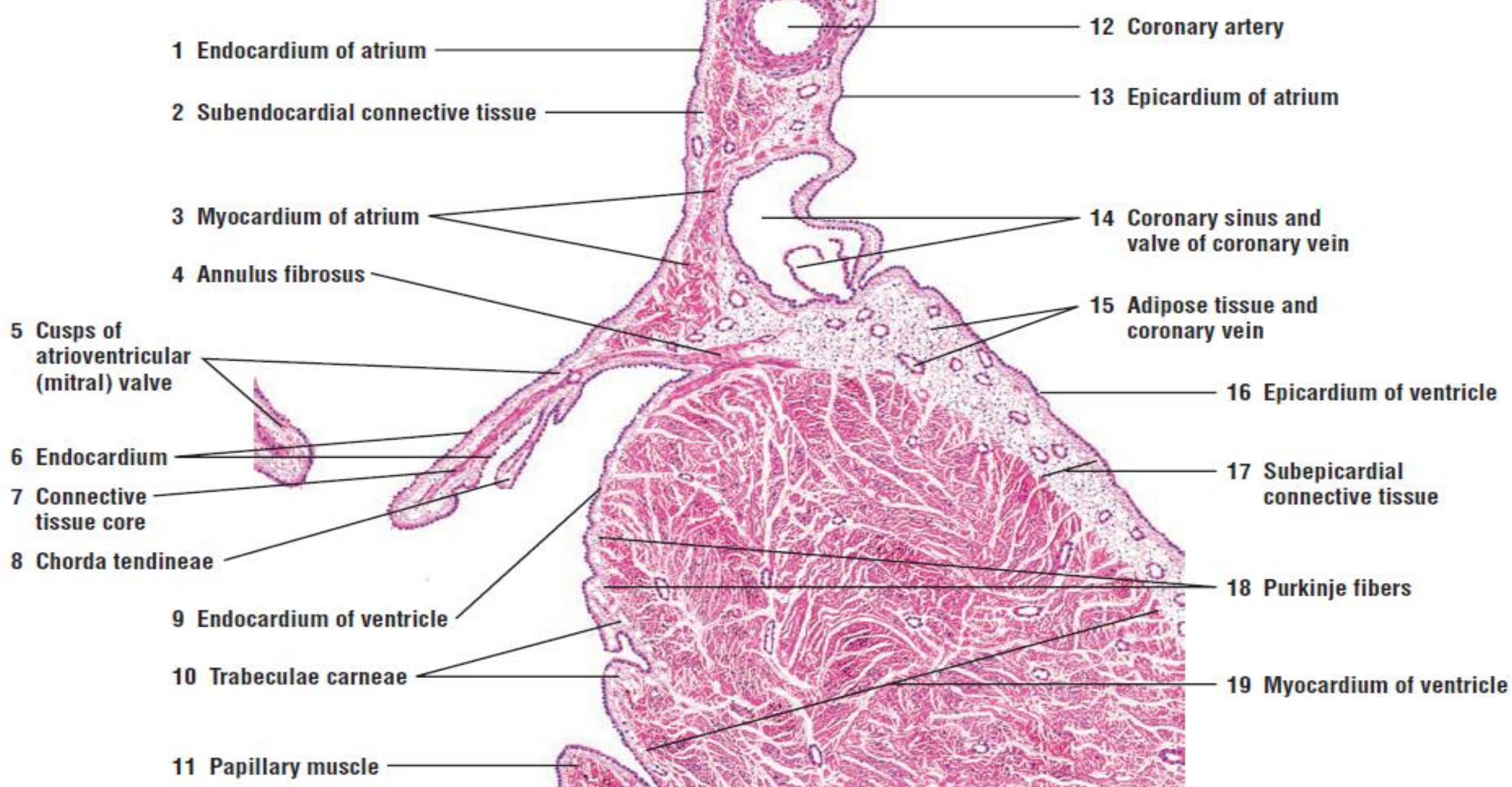


FIGURE 8.6 ■ Heart: a section of the left atrium, atrioventricular valve, and left ventricle (longitudinal section). Stain: hematoxylin and eosin. Low magnification.

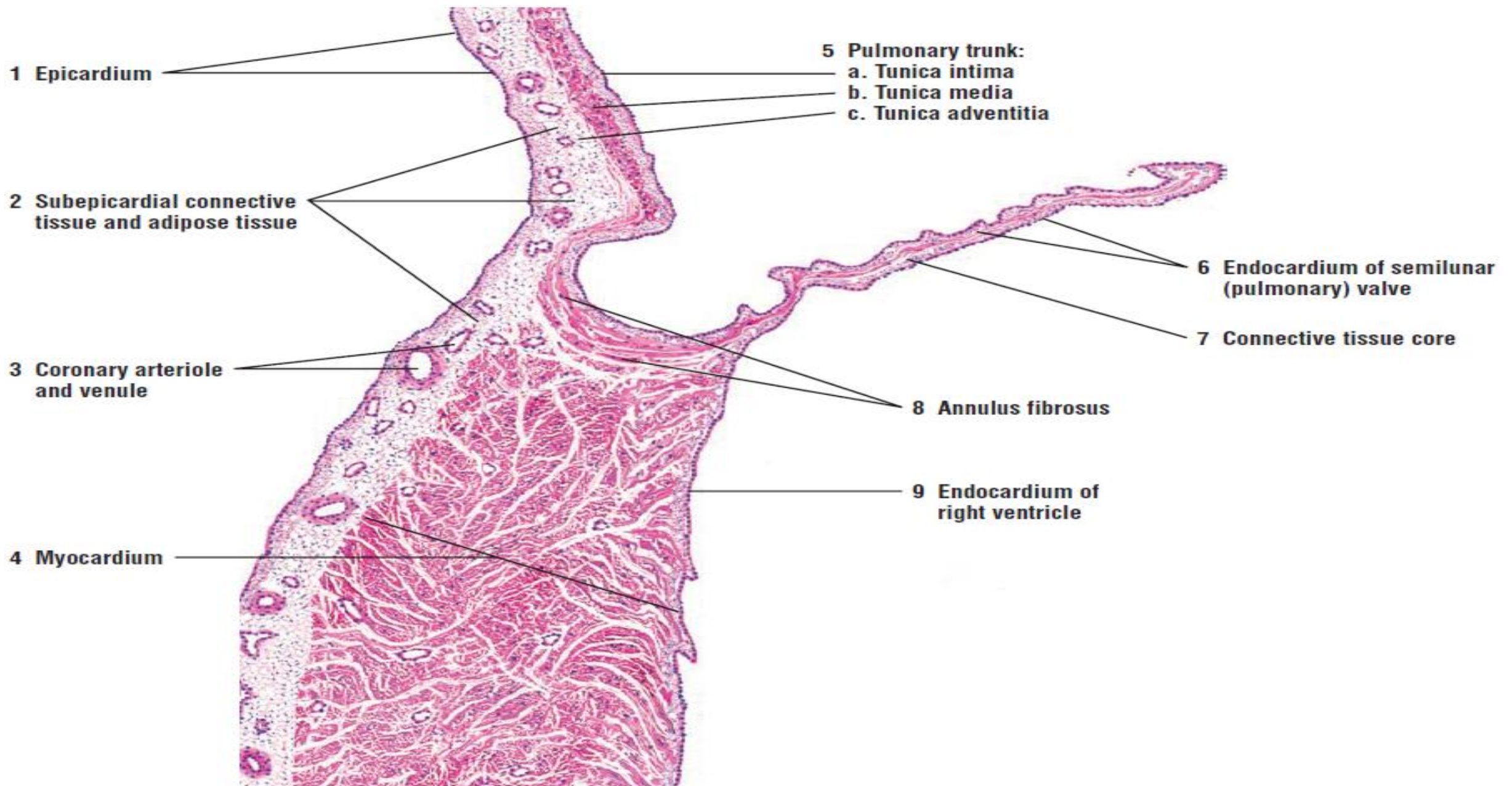


FIGURE 8.7 ■ Heart: a section of right ventricle, pulmonary trunk, and pulmonary valve (longitudinal section). Stain: hematoxylin and eosin. Low magnification.

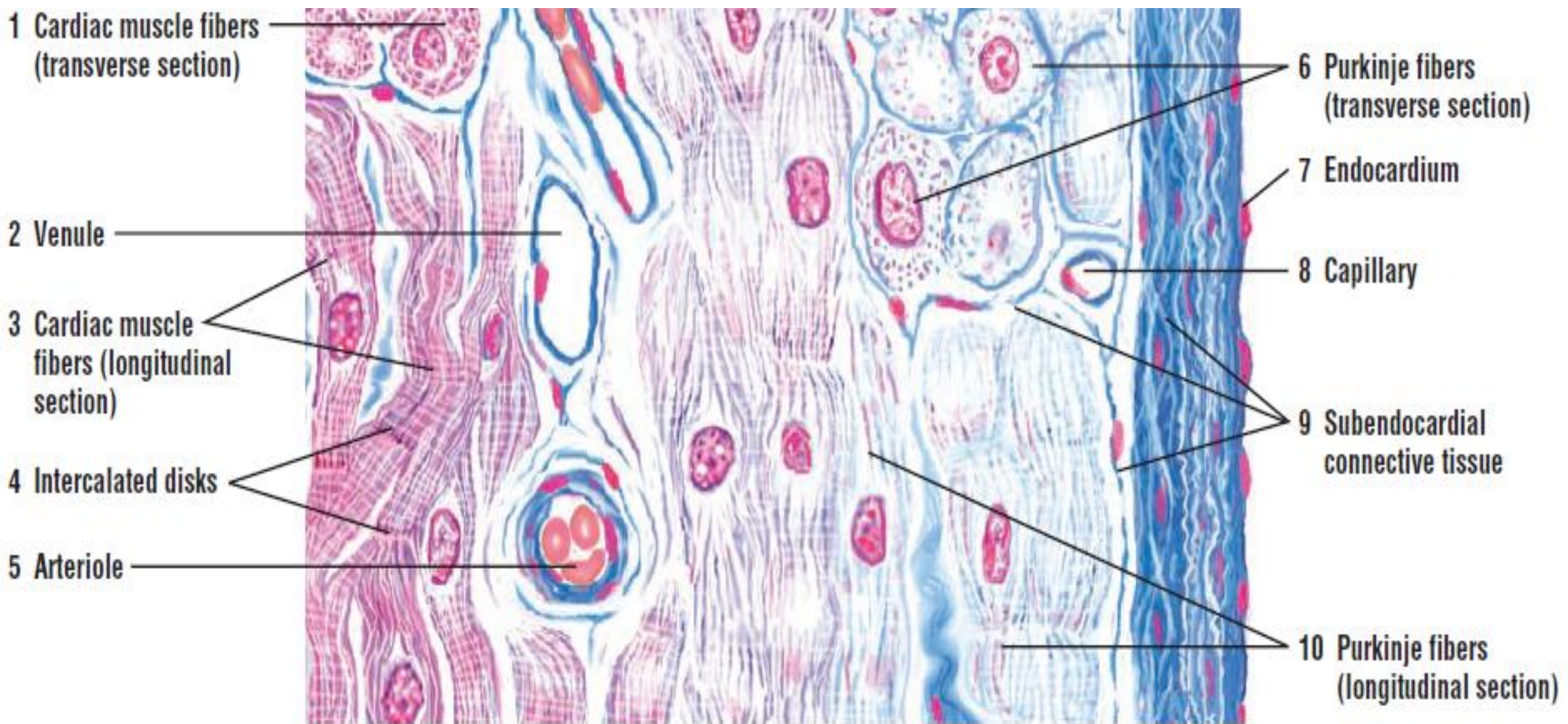
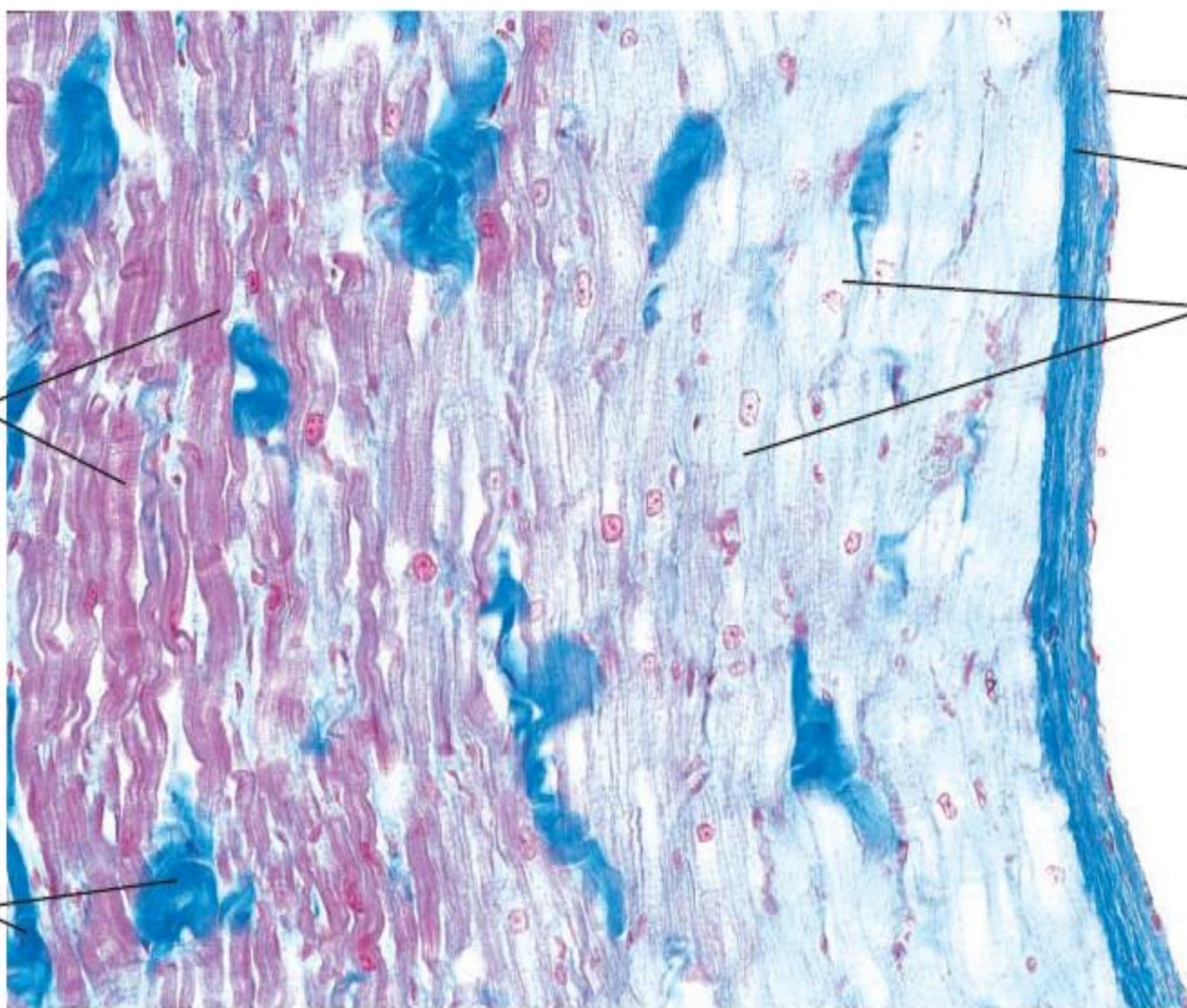


FIGURE 8.8 ■ Heart: contracting cardiac muscle fibers and impulse-conducting Purkinje fibers. Stain: Mallory-azan. High magnification.



1 Cardiac muscle fibers

2 Connective tissue fibers

3 Endocardium

4 Subendocardial connective tissue

5 Purkinje fibers

Discussion

1. What structural features make capillaries ideal for exchange between blood and tissues?
2. How does the density of the capillary plexus vary between tissues, and why?
3. What is the role of pericytes in capillary walls?
4. What are the structural differences between continuous and fenestrated capillaries?
5. In which tissues are continuous capillaries typically found, and what is the functional significance?
6. Why are fenestrated capillaries found in endocrine glands and renal glomeruli?
7. How do sinusoids differ structurally and functionally from typical capillaries?
8. What is the significance of the discontinuous or fenestrated nature of sinusoidal walls?
9. Why do some sinusoids contain phagocytic cells instead of endothelial cells?